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Unit 3 - Creating Your Campfire

PDF Documents

PDF Document 1: 1738220607543-Chapter 5 - Unit 3 - Creating Your Campfire.pdf

This PDF document is included in the unit materials.

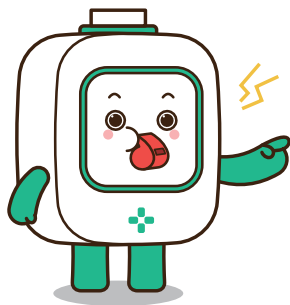
Update #1 - 2025-05-16 15:29:35



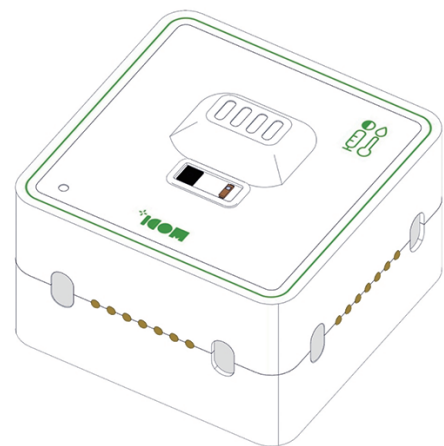
UNIT 3 CREATING YOUR CAMPFIRE

Now that you know how to set up a campfire, it's time to get creative! In this activity, we'll make a pretend campfire that you can use indoors. This is a fun way to learn about campfires without needing any real fire.

Do you remember what Environment do?



It can sense four types of things!



Environment

TEMPERATURE

Measures how hot or cold the surroundings

HUMIDITY

Checks the amount of moisture in the air.

LIGHT

Gauges the brightness of the light around it.

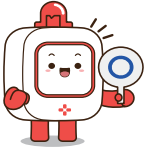
VOLUME

Detects the level of sound in the environment.

Introduction to Code Editor

What is Code Editor?

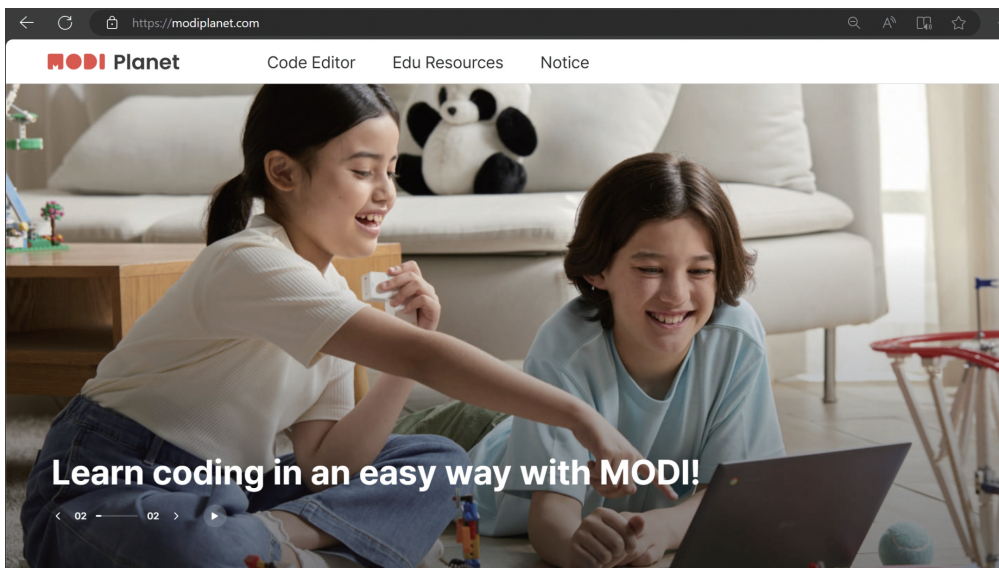
Code Editor is where we can program MODI modules with block-based programming.



Let's get started by following the next steps:

Step 1

Go to 'MODI Planet' in your web browser. Here's the link: [www.modiplanet.com](https://modiplanet.com)



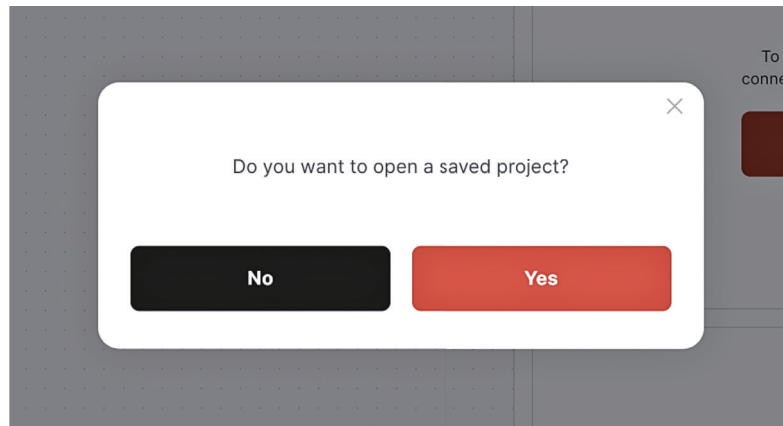
Step 2

Find 'Code Editor' in the top menu and click it.

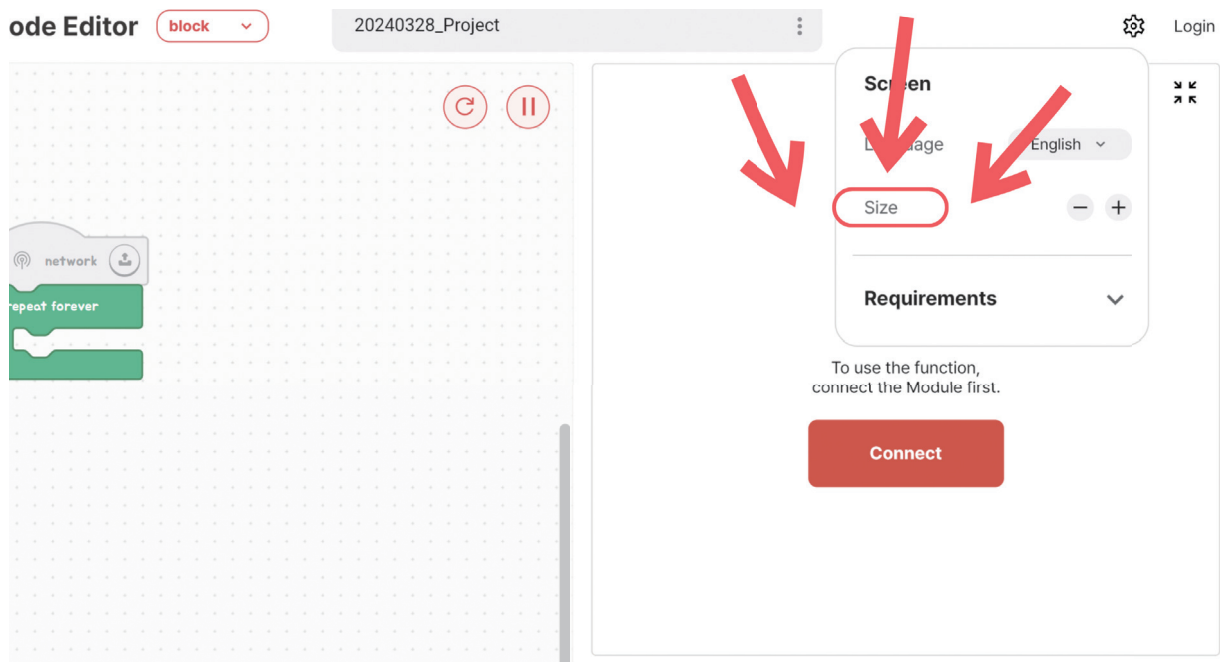




Step 3 If it asks about opening a saved project, click 'No.'

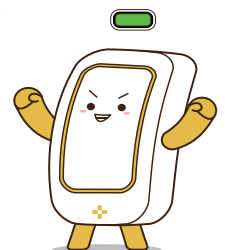


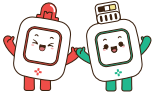
Step 4 If the screen layout doesn't suit your preference, adjust the ratio by clicking the "Set" button.



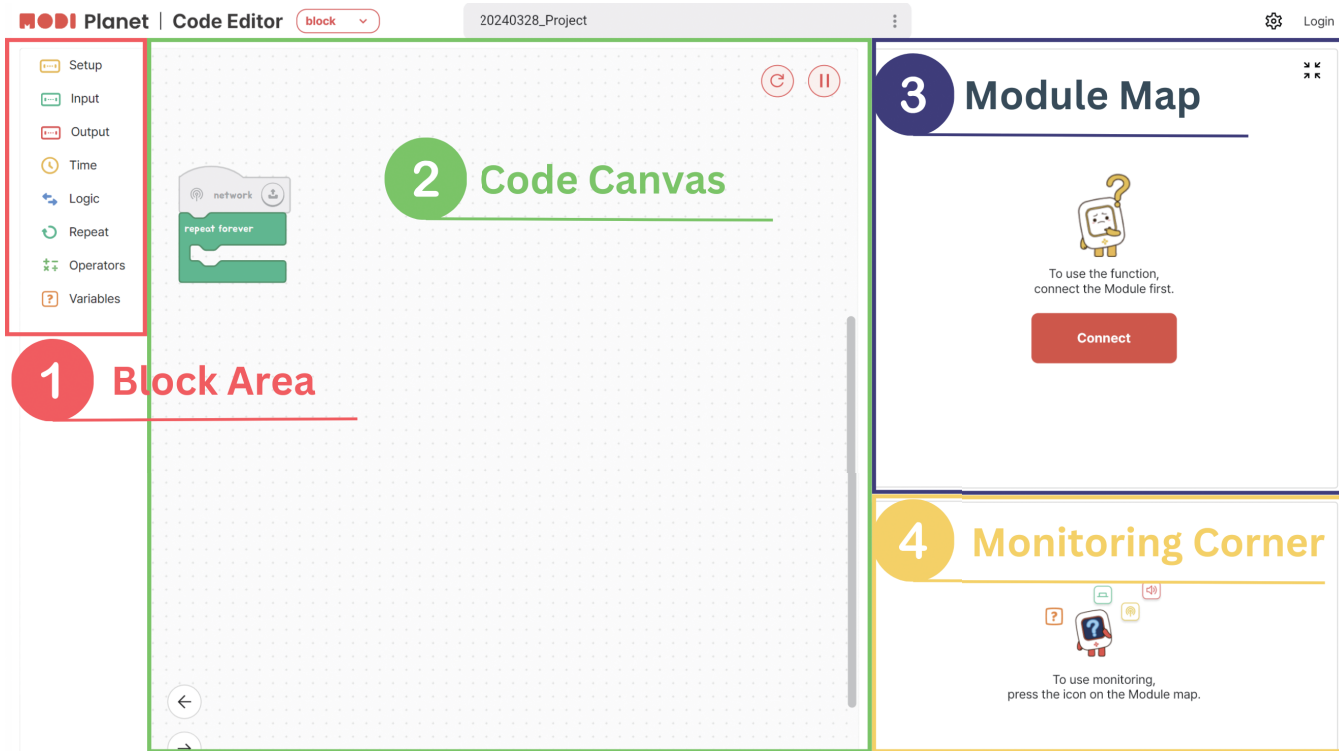
TIPS

Depending on your screen size, some options might look different. If you see a button in the top right corner, use it to see more options.





EXPLORING CODE EDITOR



1 Block Area :

This is where all the coding blocks live. They are sorted into eight groups based on what they do.

2 Code Canvas :

This is your space to build your code by putting blocks together from the Block Area.

3 Module Map :

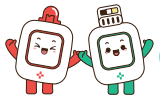
Here, you'll see icons for all the MODI modules connected to your PC. Clicking a module icon lets you check on that module.

4 Monitoring Corner :

This is where the monitoring screen appears. Depending on the size of your screen, it may appear as a pop-up window.

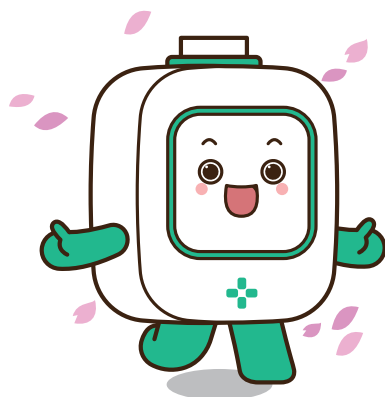
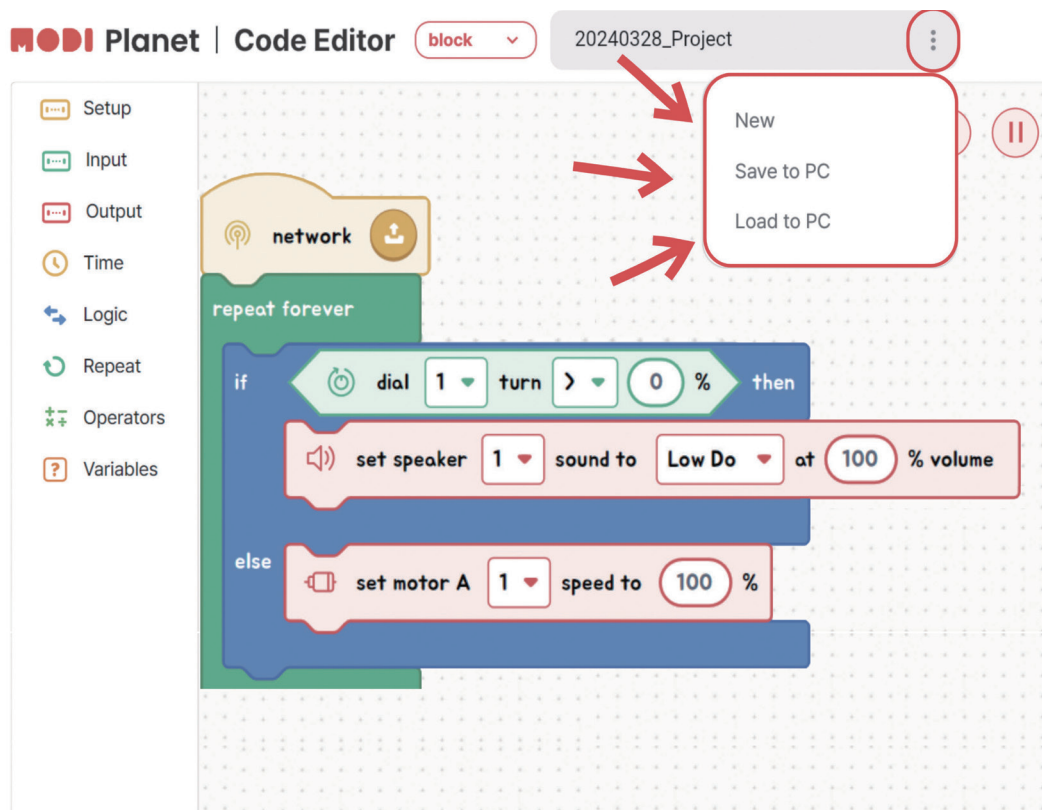


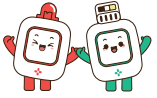
INTRODUCTION TO CODE EDITOR



SAVE & LOAD PROJECTS

Click the ellipsis in the picture below to open a popup with options to save and load your projects.





INITIALIZING MODULES

- 1 Ensure the modules you wish to initialize are connected to your device.
- 2 Click the 'Initialization' button to reset them.

MODI Planet | Code Editor block 20240328_Project

Setup
Input
Output
Time
Logic
Repeat
Operators
Variables

network

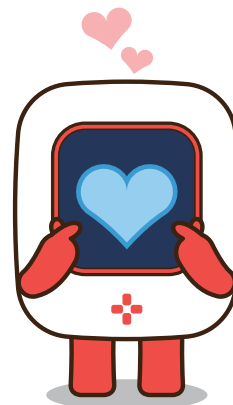
repeat forever

if dial 1 turn > 0 % then

set speaker 1 sound to Low Do at 100 % volume

else

set motor A 1 speed to 100 %



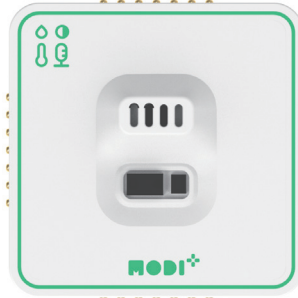


Activity: Wishing Campfire

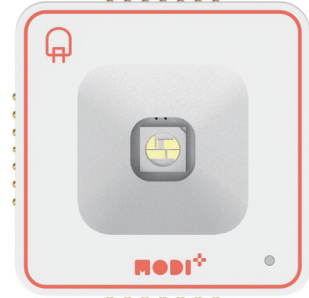
1. Materials Needed:



Network



Environment

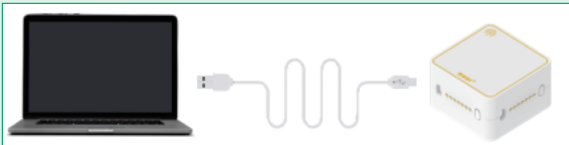


LEDs

2. Connect the Network with computer

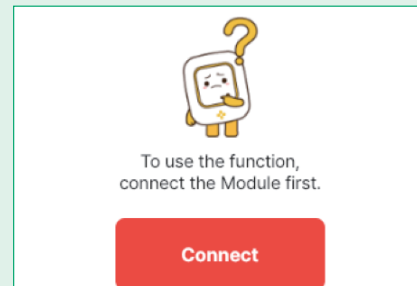
Step 1

Connect the Network Module to your device using a Type-C cable.



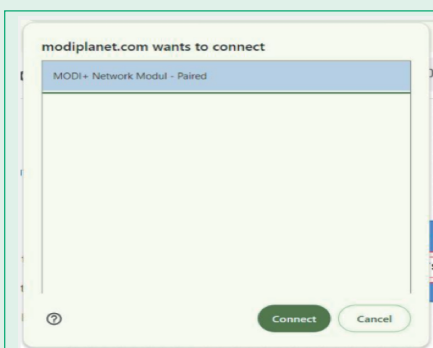
Step 2

Click the connect button in the module map.



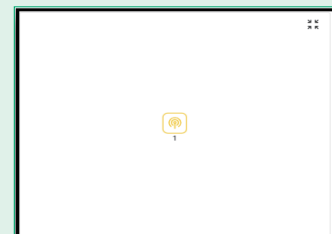
Step 3

Select 'MODI+ Network Module-Paired' and then click 'Connect' again.

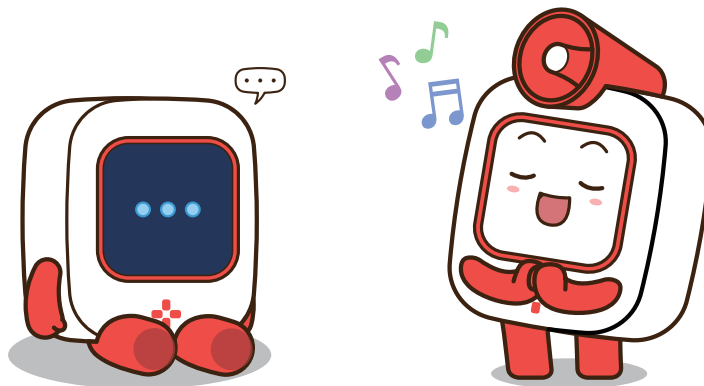


Step 4

A network icon will appear in the Module Map when the connection is successful.



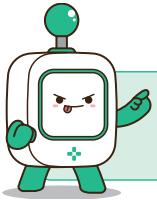
- 3.** Let's attach the Environment and LED modules next to the Network! Once everything is connected, you should see three modules displayed on the module map.





Coding

Code Story



Topic Camp Fire



DEFINITION

CODE DESIGN

We are on camp!
Let's Light it up



DECOMPOSITION

STORY TO CODE

To make the campfire flames flicker or rise by blowing, what module would be needed?
Please write down the names of the modules you can use.



ABSTRACTION

CODE DESIGN

How should it work?
Shall we write down the functions in order?



Fire is off



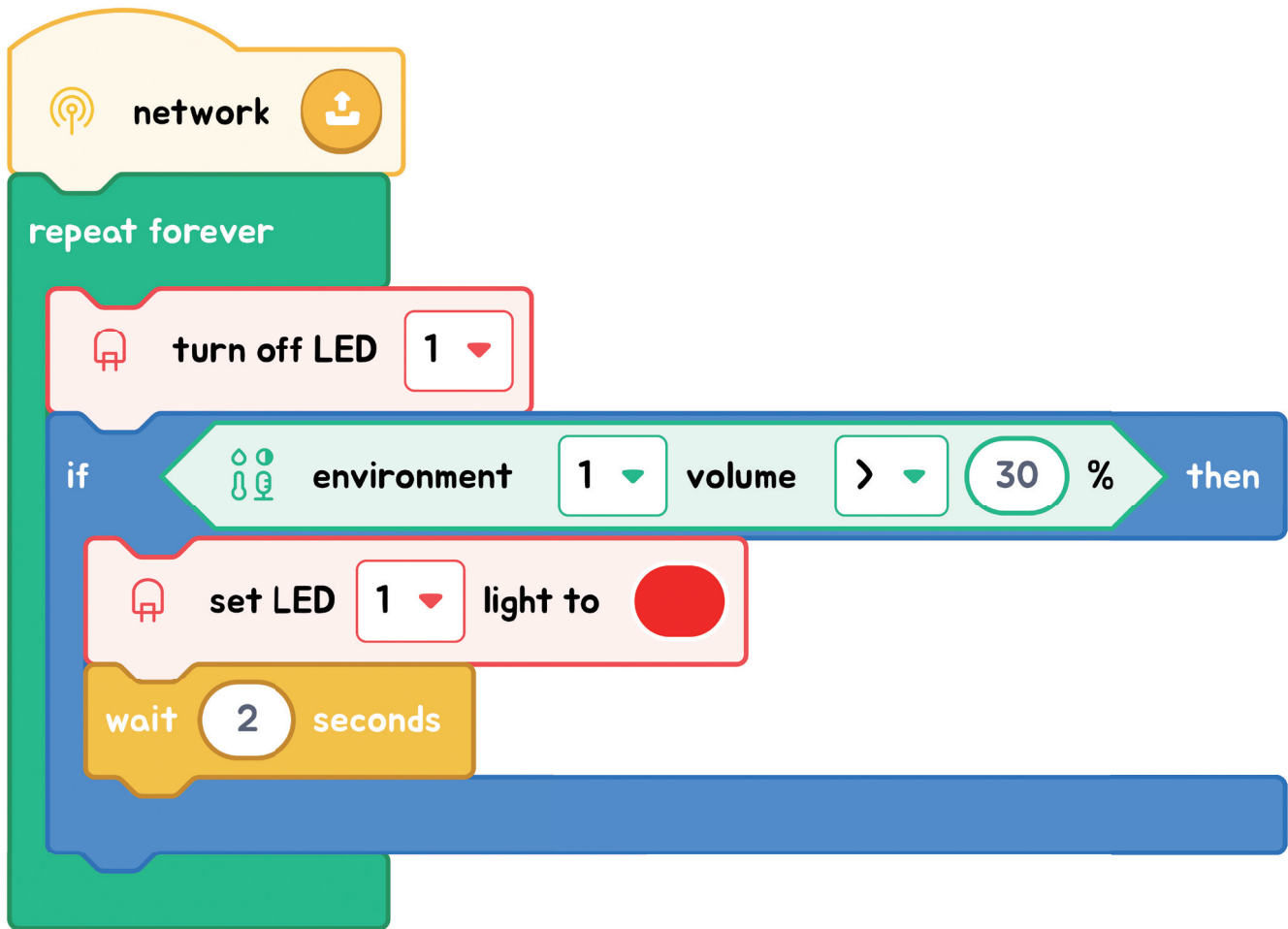
Blow the wood



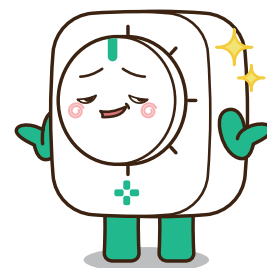
Fire is on for 2 seconds



Fire is off

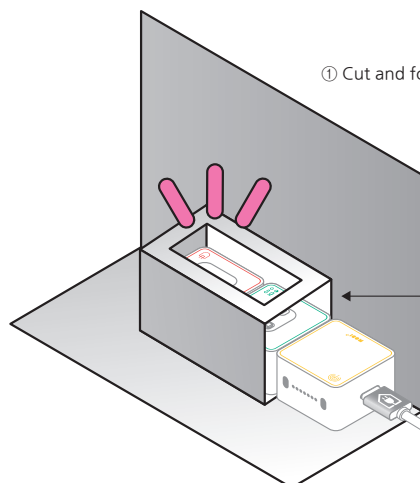
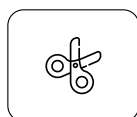


Now, let's use the template to create a camp fire!
Combine the template and the modules well to make the fire.



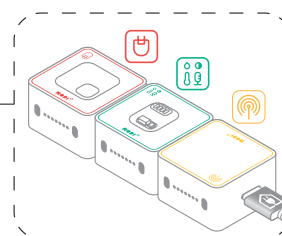


Camp Frie



① Cut and fold like a pop-up card.

② Put the modules through it.



[Guidelines]

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